Introduction to Tropical Meteorology

First-Day Warm Up Activities - Sample Answers

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Purpose of This Guide

This document provides standard or defensible explanations for the first-day activities. The goal is not to summarize one exact answer from class discussion. Rather, these notes identify the key physical ideas that should emerge from our discussion and, as we mentioned in class, some connections to future lectures.

Opening Question: What Makes the Tropics Different?

It is not geography or latitude necessarily (we will see that the tropics can support *mid-latitude like* disturbances; and conversely the mid-latitudes can support *tropics-like* disturbances). Rather, the tropics differ from the midlatitudes because several basic dynamical and thermodynamic scalings change simultaneously.

Some important points are:

- The Coriolis parameter

  $$f = 2\Omega \sin\phi$$

  decreases toward the equator and vanishes at $\phi = 0$.
- Rossby numbers are therefore often larger, so geostrophic and quasigeostrophic reasoning is less universally applicable.
- Horizontal temperature gradients are weak through much of the tropical free troposphere compared with the midlatitudes.
- Latent heating from deep convection is a leading-order component of the tropical heat budget and strongly affects the circulation.
- Moisture is dynamically active because condensation and precipitation couple water vapor to buoyancy, heating, vertical motion, and circulation.
- Large-scale tropical circulations often contain substantial divergence and vertical motion.
- The equatorial variation of f remains dynamically important even though $f = 0$ at the equator itself.

Key Message: The tropical atmosphere is not simply a watered-down (i.e., weaker gradients) version of the mid-latitude atmosphere. Yes, the gradients in temperature and pressure in the tropics are weaker than the mid-latitudes, but the interplay between moisture gradients, rotation, convection, eddies and large-scale overturning (in both zonal and meridional directions) enter the leading-order dynamics in different ways. Just like the extratropics, the tropical atmosphere can support a rich variety of waves and vortices.

Activity 1: What Is Dynamically Different About the Tropics?

A useful comparison is:

Property	**Midlatitudes**	**Tropics**
Coriolis parameter (vertical component of planetary vorticity)	Relatively large	Small near equator; changes rapidly as one moves away from the equator
Geostrophic balance	A first approximation	Less reliable for many disturbances
Thermal-Wind balance	A good approximation to account for the mean state of the extratropical atmosphere	Less reliable for many disturbances
Horizontal temperature gradients	Often strong	Usually weak in free troposphere
Baroclinicity	Central to synoptic cyclones	Generally weaker on large scales
Convection	Important but often embedded within frontal systems	Frequently a primary driver of heating and circulation
Moisture	Important thermodynamic parameter	Strongly coupled to dynamics and convection
Divergence	Small in QG scaling	Often dynamically important
Waves	Primarily baroclinic Rossby waves and gravity waves (cleanly separated)	Kelvin, equatorial Rossby, mixed Rossby–gravity, and inertia–gravity modes are ubiquitous

The phrase "most fundamental" has no single mandatory answer. Some acceptable answers are:

1. small f and the resulting change in dynamical balances;
2. weak horizontal temperature gradients;

3. strong coupling among moisture, convection, and circulation.

The above are not independent. For example, strong convective adjustment helps maintain weak free-tropospheric horizontal temperature gradients, while weak rotational constraint allows divergent overturning circulations to become dynamically prominent.

Activity 2: Does Geostrophy Work Here?

The Rossby number is

$$Ro = \frac{U}{fL}.$$

Note Use SI units and pay attention to unit conversion.

System A: Midlatitude cyclone

$$U = 15 \text{ m s}^{-1}, \qquad L = 1000 \text{ km} = 1.0 \times 10^{6} \text{ m}, \qquad f = 1.0 \times 10^{-4} \text{ s}^{-1}.$$

Therefore,

$$Ro = \frac{15}{(1.0 \times 10^{-4})(1.0 \times 10^{6})} = \frac{15}{100} = 0.15.$$

This is small enough that geostrophic balance is likely to be a useful first approximation.

System B: Tropical disturbance at $15°$N

$$U = 8 \text{ m s}^{-1}, \qquad L = 500 \text{ km} = 5.0 \times 10^{5} \text{ m}, \qquad f = 3.8 \times 10^{-5} \text{ s}^{-1}.$$

Thus,

$$Ro = \frac{8}{(3.8 \times 10^{-5})(5.0 \times 10^{5})} = \frac{8}{19} \approx 0.42.$$

Rotation remains important, but ageostrophic effects are no longer necessarily small.

System C: Disturbance near $2°$N

$$U = 8 \text{ m s}^{-1}, \qquad L = 1000 \text{ km} = 1.0 \times 10^{6} \text{ m}, \qquad f = 5.0 \times 10^{-6} \text{ s}^{-1}.$$

Thus,

$$Ro = \frac{8}{(5.0 \times 10^{-6})(1.0 \times 10^{6})} = \frac{8}{5} = 1.6.$$

Here the advective acceleration is comparable to or larger than the Coriolis acceleration, so geostrophic balance is a poor general approximation.

Answers to the Conceptual Questions

1. The approximate Rossby numbers are

$$Ro_A = 0.15, \qquad Ro_B \approx 0.42, \qquad Ro_C \approx 1.6.$$

2. System A is most nearly geostrophic.
3. System C is least geostrophic.
4. The f-plane approximation assumes f is approximately constant over the domain. Near the equator, f itself is small and changes sign across the equator, so its meridional variation cannot be neglected.
5. Earth's rotation still matters because the meridional gradient of planetary vorticity remains finite:

$$\beta = \frac{\partial f}{\partial y} \approx \frac{2\Omega}{a}$$

at the equator, where a is the Earth's radius.

Near the equator,

$$f \approx \beta y.$$

In a later module, we will see that this β effect is central to equatorial wave dynamics and permits distinct equatorially trapped modes even though $f = 0$ exactly at $y = 0$.

> **Key Message** The correct conclusion is not that rotation becomes irrelevant in the tropics. Rather, the form in which rotation enters the dynamics changes.

Activity 3: Diagnose the Tropical Atmosphere from Observations

From the figures shown, we can identify:

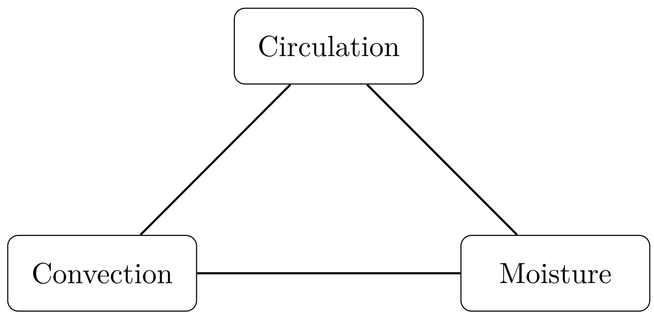
- the Intertropical Convergence Zone (ITCZ),
- subtropical highs,
- the western Pacific and Indian Ocean warm pools,
- the equatorial Pacific cold tongue,
- regions of strong deep convection and precipitation

Why Does Maximum Rainfall Not Simply Coincide with Maximum SST?

Warm sea surface temperature (SST) favors deep convection because it helps maintain a warm and moist boundary layer and enhances surface enthalpy fluxes. However, SST alone is not sufficient. Deep tropical rainfall also depends on:

- column water vapor, low-level convergence, vertical motion, convective inhibition, free-tropospheric humidity, atmospheric stability, surface fluxes, large-scale circulation, land–sea geometry and topography, ocean–atmosphere feedbacks.

A conceptual summary then can be abstracted as follows:



Rainfall is therefore an emergent product of coupled thermodynamics and dynamics rather than a local function of SST alone. For example, very warm water may coexist with suppressed convection if the large-scale environment is subsiding or dry. Conversely, strong convergence can organize intense rainfall in regions that do not possess the absolute maximum SST.

Activity 4: Tropical Meteorology Mysteries

1. Why are tropical cyclones rare within about 5° of the equator?

Tropical cyclogenesis requires the development of a rotating, approximately balanced vortex. In the simplest sense, spin-up of the tropical cyclone can be viewed as a transfer of vorticity from the planetary spin (the vertical component of planetary vorticity to be specific which is, f, the Coriolis parameter) to the storm's spin (the vertical component of relative vorticity, ζ). Near the equator, f is very small, making it difficult for weak disturbances to acquire and maintain the planetary contribution to the vorticity needed for organized cyclonic circulation. In other words, very close to the equator, the dynamical environment is usually unfavorable because the background absolute vorticity is too small.

2. Why are tropical upper-tropospheric temperatures horizontally rather uniform?

In the tropics, gravity waves and large-scale divergent circulations efficiently redistribute pressure and temperature anomalies. Deep convection also tends to adjust temperature profiles toward moist-adiabatic structures. This contributes to the weak-temperature-gradient character of the tropical free troposphere. A useful scaling idea is that large horizontal temperature gradients are difficult to maintain when rotation is weak and gravity-wave adjustment is rapid.

3. Why are easterly winds common in the tropical lower troposphere?

The trade winds arise as part of the large-scale Hadley circulation and subtropical pressure distribution. Near-surface air moves equatorward from the subtropical highs and is deflected westward by the Coriolis force, producing the northeast trades in the Northern Hemisphere and southeast trades in the Southern Hemisphere.

4. Why is the ITCZ often displaced away from the equator?

The latitude of the ITCZ reflects the latitude of strongest low-level convergence and deep ascent. Its position is affected by several factors, including:

- hemispheric differences in radiative heating,
- ocean circulation,
- land distribution,
- cross-equatorial energy transport,
- SST gradients,
- seasonal migration of solar heating.

The ITCZ therefore need not be centered on the geographic equator.

5. Why can tropical rainfall vary greatly while horizontal temperature gradients remain weak?

Unlike the extratropics, where quasigeostrophic dynamics provide a useful framework for diagnosing large-scale ascent and thereby the environments favorable for precipitation, tropical rainfall is strongly coupled to the evolution of moisture and the large-scale circulation. Tropical moisture is controlled by surface evaporation, horizontal and vertical advection, and cloud and precipitation processes. At the same time, gravity-wave adjustment efficiently removes large-scale free-tropospheric temperature anomalies, producing the weak-temperature-gradient character of much of the tropical atmosphere. Consequently, tropical variability can be pronounced in moisture, convection, precipitation, and circulation even when free-tropospheric temperature variations remain relatively small.

6. How can waves exist when horizontal temperature gradients are weak?

A background horizontal temperature gradient is not required for all atmospheric waves. As we will see later in the course, the tropical atmosphere supports a wide range of wave motions. The equatorial β effect provides restoring dynamics and equatorial trapping. Moist convection can then couple to these dry dynamical modes to produce convectively coupled equatorial waves.

8. Why are warm oceans necessary but not sufficient for tropical cyclogenesis?

Warm SST supports large surface enthalpy fluxes and a moist convective boundary layer. However, tropical cyclogenesis also requires favorable environmental and dynamical conditions, commonly including: sufficient background vorticity, a pre-existing disturbance, a deep saturated column of air, sustained deep convection, relatively weak vertical wind shear, and sufficient oceanic energy supply.

Thus,

$$\text{warm SST} \neq \text{automatic tropical cyclogenesis.}$$

Overall First-Day Takeaway

1. Familiar dynamical principles still apply in the tropics, but their relative importance changes.
2. Moisture, radiative effects (we have barely discussed this in this activity!) and convection are not peripheral corrections to dynamics and behavior of weather systems in the tropics; they are central components of tropical atmospheric dynamics.
3. Tropical meteorology is fundamentally a multiscale problem involving interactions among radiation, convection, circulation, waves, and the ocean.